

8.4.5

EE25BTECH11002 - Achat Parth Kalpesh

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Question

An ellipse is drawn by taking a diameter of the circle $(x - 1)^2 + y^2 = 1$ as its semi minor axis and a diameter of the circle $x^2 + (y - 2)^2 = 4$ as semi-major axis. If the centre of the ellipse is at the origin and its axes are the coordinate axes, then the equation of the ellipse is

① $4x^2 + y^2 = 4$

② $x^2 + 4y^2 = 8$

③ $4x^2 + y^2 = 8$

④ $x^2 + 4y^2 = 1$

Solution

The standard equation of a circle is given as

$$(\mathbf{x} - \mathbf{c})^\top (\mathbf{x} - \mathbf{c}) = r^2 \quad (1)$$

Given two circles are

$$(\mathbf{x} - \mathbf{c}_1)^\top (\mathbf{x} - \mathbf{c}_1) = 1 \quad (2)$$

$$(\mathbf{x} - \mathbf{c}_2)^\top (\mathbf{x} - \mathbf{c}_2) = 4 \quad (3)$$

The centers and radii are

$$\mathbf{c}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad r_1 = 1 \quad (4)$$

$$\mathbf{c}_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \quad r_2 = 2 \quad (5)$$

Solution

Verifying that the origin lies on both circles:

$$(\mathbf{0} - \mathbf{c}_1)^\top (\mathbf{0} - \mathbf{c}_1) = 1 = r_1^2 \quad (6)$$

$$(\mathbf{0} - \mathbf{c}_2)^\top (\mathbf{0} - \mathbf{c}_2) = 4 = r_2^2 \quad (7)$$

Thus, the diameters of both circles passing through the origin are along the directions

$$\mathbf{c}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (\text{along X-axis}) \quad (8)$$

$$\mathbf{c}_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad (\text{along Y-axis}) \quad (9)$$

Solution

Each circle's diameter length is $2r$. Therefore, the ellipse's semi-axes are equal to the respective radii:

$$b = r_1 = 1 \quad (\text{semi-minor axis}) \quad (10)$$

$$a = r_2 = 2 \quad (\text{semi-major axis}) \quad (11)$$

The standard equation of an ellipse centered at the origin with coordinate axes as its axes is

$$\mathbf{x}^\top A \mathbf{x} = 1 \quad (12)$$

where

$$A = \begin{pmatrix} \frac{1}{b^2} & 0 \\ 0 & \frac{1}{a^2} \end{pmatrix} \quad (13)$$

Solution

Substituting $a = 2$, $b = 1$,

$$A = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \quad (14)$$

Hence,

$$\mathbf{x}^\top \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \mathbf{x} = 1 \quad (15)$$

Multiplying throughout by 4 gives

$$\mathbf{x}^\top \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} = 4 \quad (16)$$

or equivalently,

$$4x^2 + y^2 = 4 \quad (17)$$

```
#include <stdio.h>
#include <math.h>

#define PI 3.14159265358979323846

// Function to generate ellipse points
void generate_ellipse(double a, double b, int n, double x[],
    double y[]) {
    for(int i = 0; i < n; i++) {
        double theta = 2 * PI * i / n;
        x[i] = b * cos(theta);
        y[i] = a * sin(theta);
    }
}
```

```
// Function to generate circle points
void generate_circle(double cx, double cy, double r, int n,
    double x[], double y[]) {
    for(int i = 0; i < n; i++) {
        double theta = 2 * PI * i / n;
        x[i] = cx + r * cos(theta);
        y[i] = cy + r * sin(theta);
    }
}
```


Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from ctypes import CDLL, c_double, c_int, POINTER

# Load the shared library
formula = CDLL('./formula.so')

# Number of points
n = 400
arr_type = c_double * n

# Prepare arrays
x_ellipse = arr_type()
y_ellipse = arr_type()
x_circle1 = arr_type()
```

Python Code

```
y_circle1 = arr_type()
x_circle2 = arr_type()
y_circle2 = arr_type()

# Call C functions
formula.generate_ellipse(c_double(2), c_double(1), c_int(n),
    x_ellipse, y_ellipse)
formula.generate_circle(c_double(0), c_double(2), c_double(2),
    c_int(n), x_circle1, y_circle1)
formula.generate_circle(c_double(1), c_double(0), c_double(1),
    c_int(n), x_circle2, y_circle2)

# Convert to numpy arrays
x_ellipse = np.array(x_ellipse)
y_ellipse = np.array(y_ellipse)
x_circle1 = np.array(x_circle1)
y_circle1 = np.array(y_circle1)
x_circle2 = np.array(x_circle2)
y_circle2 = np.array(y_circle2)
```

Plotting

```
plt.figure(figsize=(10,10))
plt.plot(x_ellipse, y_ellipse, 'g', linewidth=2, label=r'Ellipse:
    $4x^2 + y^2 = 4$')
plt.plot(x_circle1, y_circle1, 'b', linewidth=2, label=r'Circle:
    $C_1$ (0,2), r=2')
plt.plot(x_circle2, y_circle2, 'r', linewidth=2, label=r'Circle:
    $C_2$ (1,0), r=1')
```

Mark centers

```
plt.plot(0, 2, 'bo')
plt.text(0.1, 2.1, '$C_1$(0,2)', color='blue', fontsize=12)
plt.plot(1, 0, 'ro')
plt.text(1.1, 0.1, '$C_2$(1,0)', color='red', fontsize=12)
```

```
# Axes and formatting
plt.axhline(0, color='k', linewidth=0.8)
plt.axvline(0, color='k', linewidth=0.8)
plt.gca().set_aspect('equal', adjustable='box')
plt.title('Ellipse and Circles with Centers')
plt.xlabel('X - axis')
plt.ylabel('Y - axis')
plt.legend(loc='upper right')
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()
```

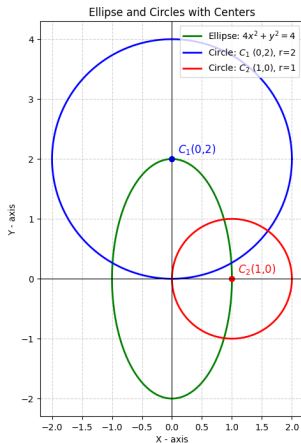


Figure: Ellipse and Circles